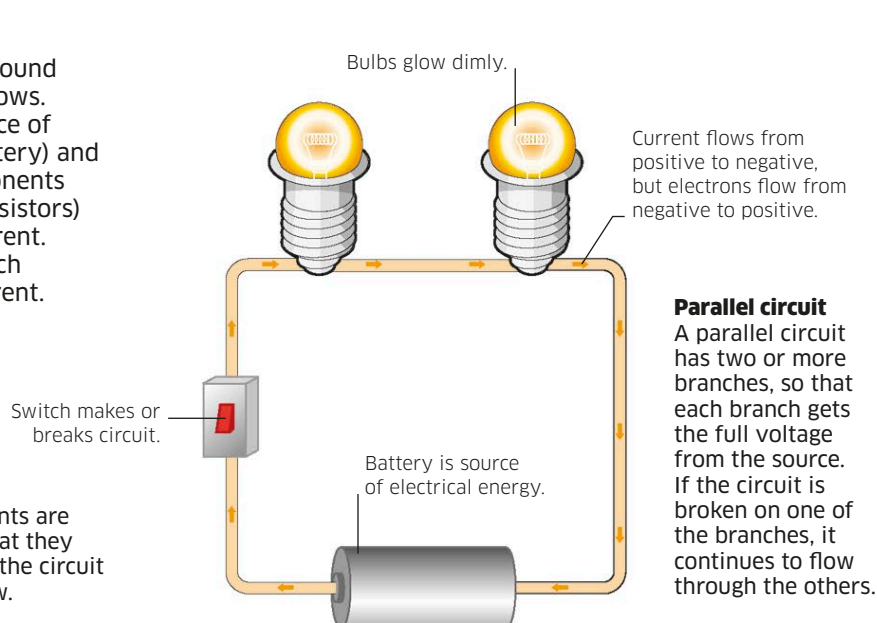


Electric circuits

An electric circuit is the path around which a current of electricity flows. A simple circuit includes a source of electrical energy (such as a battery) and conducting wires linking components (such as switches, bulbs, and resistors) that control the flow of the current. Resistance is the degree to which materials resist the flow of current.

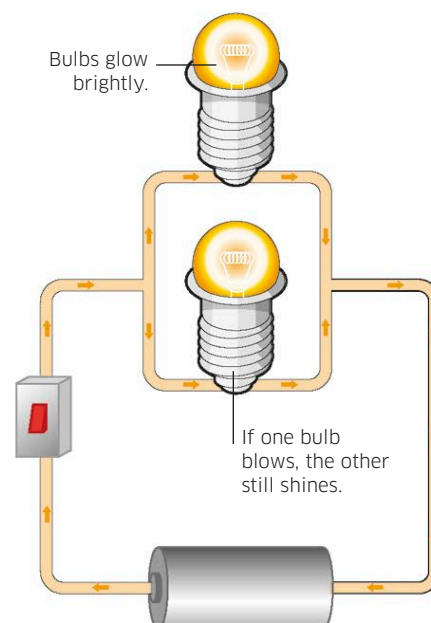
Series circuit

In a series circuit, all the components are connected one after another, so that they share the voltage of the source. If the circuit is broken, electricity ceases to flow.



Parallel circuit

A parallel circuit has two or more branches, so that each branch gets the full voltage from the source. If the circuit is broken on one of the branches, it continues to flow through the others.

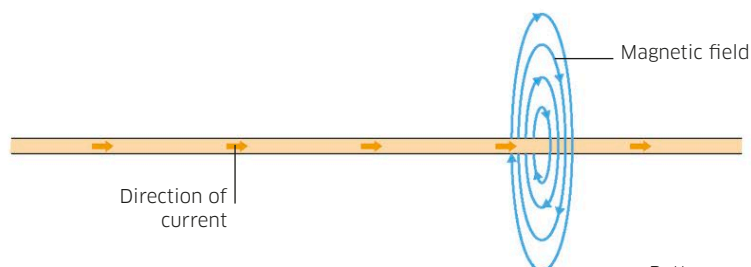


Electromagnetism

Moving a wire in a magnetic field causes a current to flow through the wire, while an electric current flowing through a wire generates a magnetic field around the wire. This creates an electromagnet—a useful device because its magnetism can be switched on and off.

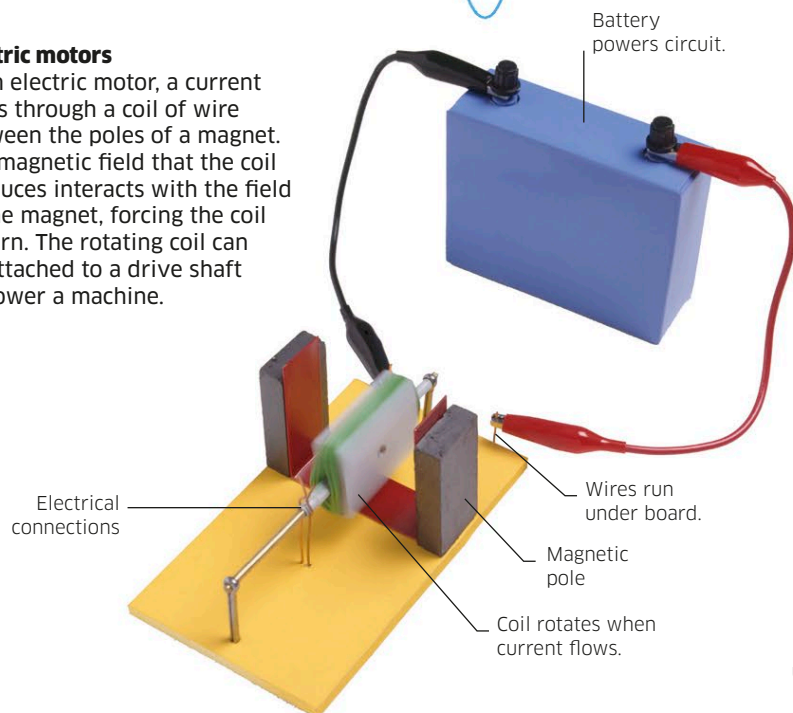
Electromagnetic field

When an electric current flows through a wire, it generates rings of magnetic field lines all around it. You can see this by placing a compass near a wire carrying a current. The stronger the current, the stronger the magnetism.



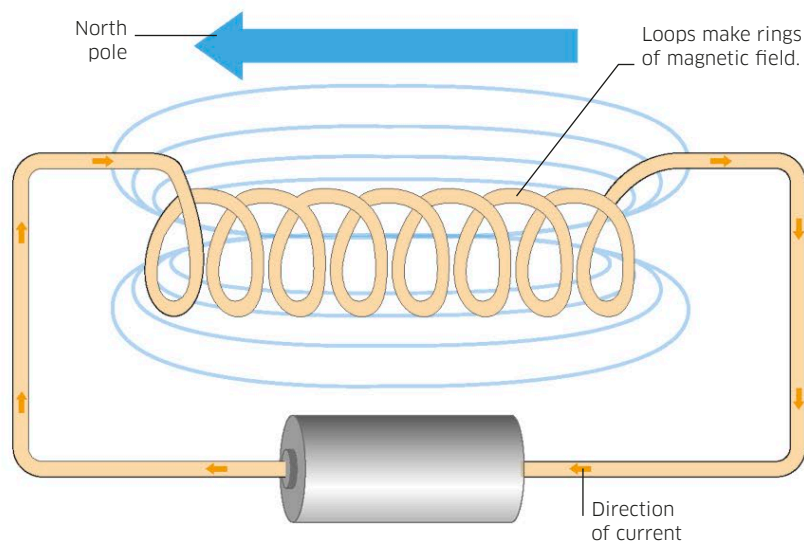
Electric motors

In an electric motor, a current flows through a coil of wire between the poles of a magnet. The magnetic field that the coil produces interacts with the field of the magnet, forcing the coil to turn. The rotating coil can be attached to a drive shaft to power a machine.



Solenoid

A coil of wire carrying a current produces a stronger magnetic field than a straight wire. This coil creates a common type of electromagnet called a solenoid. Winding a solenoid around an iron core creates an even more powerful magnetic field.



Electric generators

In a generator, a current is produced by rotating a wire coil between a magnet's poles, or by rotating a magnet while the coil is static. Generators can be big enough to power a city, or small, portable devices for supplying electricity to individuals.

